Visual Intelligence

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CIHM: context-insight hybrid Mamba for efficient medical image segmentation

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Abstract

In medical image segmentation, U-Net and its variants are widely utilized. However, CNN-based models are limited in long-range dependency modeling, while Transformer-based models are constrained by quadratic computational complexity. Recently, state space models (SSMs) such as Mamba have emerged as a promising solution, as they excel at modeling long-range interactions while maintaining linear computational complexity. This paper proposes the CIHM, a hybrid Mamba model that provides contextual insights. Its key component is the context-insight Mamba-CNN layer (CiMC layer). The CiMC layer adopts a hybrid Mamba-CNN approach to process features across different scales and levels in parallel, thereby enhancing the semantic segmentation of medical images. Specifically, the SSM branch of this layer captures long-range dependencies with comprehensive contextual semantic understanding, while the CNN block achieves superior local feature extraction using T-shaped convolution that focuses on patch centers. Moreover, to further aggregate multi-scale contextual information of medical images, we design the multiscale refining detail bridge (MRDB) module. This module significantly improves feature representation through the dense multiplication and concatenation of features for fusion. Comprehensive experiments on the ISIC2017, ISIC2018, Synapse, and DSB18 datasets demonstrate that CIHM exhibits strong competitive performance and efficiency in medical image segmentation tasks. Notably, CIHM achieves better segmentation performance than the well-known U-Net and the modern hybrid network U-Mamba, while reducing the parameter count by 40 times and 345 times, respectively. This demonstrates the potential of our approach to advance model lightweighting.

Keywords: Context-insight, Hybrid Mamba network, Lightweight, Medical image segmentation, State space model

1 Introduction

As computer technology and hardware computational capabilities continue to advance, computer-assisted diagnosis has become widely applied in the medical field, with medical image segmentation being a crucial component. In recent years, deep learning networks represented by convolutional and Transformer architectures have shown significant performance in the domain of medical image segmentation. Convolution excels in local feature extraction but is limited by its local receptive field [1], making it inadequate for capturing global context and long-term dependencies [2, 3]. Even with large-kernel convolutions [4] to handle remote information, the results remain unsatisfactory. Transformer-based models can extract remote information using sequences of contiguous patches, demonstrating superior performance in global modeling. However, self-attention mechanism requires quadratic complexity with respect to image size, resulting in an excessive computational burden [5], especially for tasks requiring dense predictions like med-

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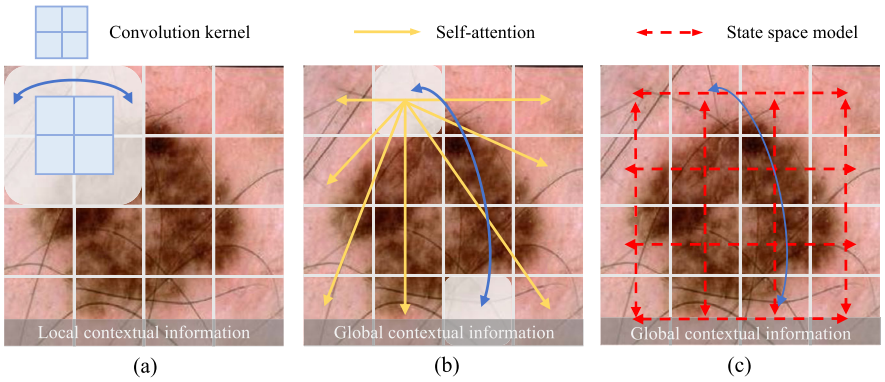


Figure 1 Illustration of different architectures for processing medical images. (a) and (b) show that CNN-based models and Transformer-based models may have difficulty capturing the global context or may be computationally complex. (c) illustrates that the Mamba architecture is well suited for medical image processing because of its efficient global contextual modeling capabilities

ical image segmentation. As illustrated in Fig. 1(a) and Fig. 1(b), they may struggle to fully capture global context or may be computationally complex. Furthermore, low parameter count and minimal computational overhead are important considerations for mobile medical segmentation tasks in clinical and real-world medical environments [6, 7]. These limitations of current models have driven us to develop a novel medical image segmentation architecture that can effectively capture remote information while maintaining linear computational complexity.

Recently, state space modeling (SSM) has garnered widespread attention. Building on the foundations laid by classical SSM research [8], modern SSMs such as Mamba [9] not only exhibit linear complexity with respect to input size and memory usage [10, 11], but they are also adept at capturing long-range dependencies. This allows them to effectively address the challenges of convolution for long-distance information extraction. Additionally, SSM-based models have been extensively researched in various fields, including language understanding [12] and general vision [10]. Considering the efficient global contextual modeling capabilities, the Mamba architecture is well-suited for medical image processing, as illustrated in Fig. 1(c). Contemporary efforts like U-Mamba [13] have introduced a model that incorporates Mamba blocks based on nnUNet, marking its first application in medical image segmentation tasks. However, it introduced a significant computational load (67M parameters and 115G FLOPs), posing challenges for deployment in mobile medical settings. This limitation highlights a critical gap, namely the lack of lightweight architectures that can effectively balance the high computational cost of Mamba with the need for precise medical image segmentation.

Although these models achieved effective results in various medical image segmentation tasks, the potential of Mamba has not yet been fully explored, and inherent challenges in medical image segmentation remain unresolved, which serve as the core motivations for this work, primarily in three aspects. First, current methods seldom consider the comprehensive integration of local and global information in medical images, which may limit the capability of the CNN-provided features in their models [14]. Second, thanks to Mamba's linear complexity, its attribute of greater spatial flexibility has been overlooked [15], which also suggests that Mamba could enable models to have varying levels of spatial modeling capabilities. Lastly, there is a significant discrepancy between different types of lesions in various medical image datasets. Lesions have shapes and colors similar to surrounding tissues, which makes them difficult to distinguish due to lower contrast. This makes it easy for small lesions to be submerged in a complex background. However, existing networks struggle to precisely define boundary features due to inadequate integration of fine-grained local details and global contextual information. This constitutes a key bottleneck for further performance improvements in biomedical image segmentation.

This paper introduces a context-insight hybrid Mamba model, called CIHM. It is an effective Mamba-CNN U-Net architecture, which incorporates three key components: the context-insight Mamba-CNN layer (CiMC layer), the T-shaped convolution (TConv) block, and the multiscale refining detail bridge (MRDB) module. These components collectively address the aforementioned challenges at multiple levels. In particular, as the core components of the encoder and decoder, the TConv block extracts more refined local detail features, while the CiMC layer adopts a duplex gated fusion structure to dynamically adjust the interaction and aggregation of global and local features. CIHM

also integrates the vision-selective selective scan mechanism (V^2SSM), which efficiently captures comprehensive global information and achieves vision-specific adaptability. The MRDB module enhances the skip connections in U-Net by aggregating multi-scale semantic feature information through feature multiplication and concatenation, reducing interference from shallow feature noise and improving feature representation. The synergistic integration of these modules enables us to more effectively tackle the challenges in medical image segmentation, and the proposed CiMC layer achieves excellent performance with extremely low parameters. Our contributions are summarized as follows:

1) We introduce CIHM, a U-Net architecture that effectively combines the Mamba and CNN frameworks for medical image segmentation. It efficiently integrates local-global information while providing comprehensive understanding of semantic context. Validation on four publicly available medical image segmentation datasets demonstrates CIHM's highly competitive performance.
2) We propose a hybrid module called the CiMC layer, which comprehensively understands semantic contextual information, effectively extracts and integrates global-local information, and minimizes redundant information and computational burden through its innovative design.
3) We introduce the MRDB module to enhance skip connections by aggregating multi-scale semantic feature information, reducing noise interference on shallow features, and thereby enabling the capture of more boundary details in shallow information.
4) The proposed method incorporates a lightweight hybrid Mamba model (0.194M parameters, 0.182G FLOPs). Furthermore, compared to the currently outstanding lightweight model LightM-UNet [16], CIHM reduces parameters by 52% and computational costs by 54%. This is of great significance for effectively addressing the challenges posed by computational resource constraints in real-world medical scenarios.

2 Related work

2.1 Medical image segmentation methods

Semantic segmentation plays a crucial role in medical image analysis as it is essential for examining anatomical structures [17]. In recent years, driven by deep learning approaches, medical image segmentation has made significant progress. Since the introduction of the U-Net design [18], it has been widely used for medical image segmentation, achieving state-of-the-art results in various medical image segmentation tasks [19–23]. This success is attributed to the skip connection paths in U-Net, which effectively preserve and fuse high-level and low-level features. This is crucial for medical image segmentation requiring fine details. Moreover, Attention U-Net [24] enhances segmentation accuracy by integrating the attention gate module into the skip connections, thereby improving the model's ability to focus on target regions. However, due to the local receptive field of convolution operations [25, 26], CNN-based techniques have limitations in learning long-range dependencies.

The Transformer is renowned for its powerful ability to model global dependencies through the core self-attention mechanism, and Vision Transformer (ViT) [5] has effectively applied this capability to image processing tasks. In contrast, CNNs show advantages in capturing local features. The combination of these two paradigms helps to fully leverage both local and global information, thereby improving segmentation accuracy. For instance, TransUNet [27] improves medical image segmentation by adding Transformer layers at the bottom of the CNN encoder for global context modeling. UNETR [28] uses Transformers to capture global multi-scale information and then feeds these features into the CNN decoder, though it leads to quadratic complexity and larger model sizes. Swin UNETR [29] addresses this issue by integrating Swin Transformer [30] as the encoder and convolution operations as the decoder, reducing model complexity to linear. Nevertheless, Swin UNETR faces challenges in computational load and efficiency due to numerous cyclic shifting operations. Additionally, TransSea [19] resolves the intrinsic semantic differences between local and global features by introducing a semantic integration module, effectively fusing convolution and Transformers seamlessly.

2.2 State space models (SSM)

Recently, Gu and Dao [9] proposed a data-related state space models (SSM) layer and constructed a general language model backbone, Mamba, which outperforms Transformers of various sizes on large-scale real-world data and has linear scaling in sequence length. With the introduction of Vision Mamba [11] and Vmamba [10], the SSM-based Mamba architecture is emerging in the field of image tasks [10]. To adapt to medical image segmentation tasks, both U-Mamba [13] and VM-UNet [31] incorporate SSM into the U-Net architecture to enhance the ability to capture long-range dependencies. At the same time, Seg-Mamba [32] combines the U-shape structure with Mamba to model the entire global features at various scales. Swin-UMamba [14] reveals how to enhance Mamba's performance in medical image segmentation using ImageNet-based training. In addition, LightM-UNet [16] utilizes Vision Mamba's residuals to further extract deep semantics and long-range relationships, achieving better performance with fewer parameters.

2.3 Lightweight frameworks for resource-limited devices

Lightweight frameworks are designed to balance accuracy and efficiency on resource-constrained hardware. Lightweight frameworks have been increasingly studied to reduce computational costs and model sizes, enabling efficient image segmentation tasks. Some early decomposition-based approaches, such as MobileNets [33], reduce parameters and FLOPs through pointwise convolution and grouped convolution. ShuffleNet [34] uses channel shuffling to overcome the performance degradation caused by grouped convolution, while ShuffleNetV2 [35] considers both memory access cost and FLOPs to achieve a trade-off between speed and accuracy. Singh et al. [36] proposed a lightweight model based on parallel dilated convolution and self-attention for medical image segmentation. Although these models have fewer parameters, they still face performance gaps and limitations in generalization capability compared to typical segmentation networks like U-Net [18], particularly when adapting to different datasets. Developing a model with high segmentation performance tailored to different resource-constrained medical tasks is crucial.

3 The proposed method

3.1 Preliminaries

3.1.1 State space models (*S4*)

State space models (SSMs) are a general family of sequence models used in deep learning that are influenced by systems capable of mapping one-dimensional sequences in a continuous manner. These models transform an input D-dimensional sequence $x(t) \in \mathbb{R}^{L\times D}$, where t denotes the continuous time step and L represents the overall sequence length, into an output sequence $y(t) \in \mathbb{R}^{L\times D}$. This transformation is achieved by utilizing a learnable latent state $h(t) \in \mathbb{R}^{N\times D}$, which encodes the hidden system dynamics at step t with a state dimension of N, and is not directly observable. The mapping process can be formulated using Eq. (1):

$$\begin{aligned} h'(t) &= \boldsymbol{A}h(t) + \boldsymbol{B}x(t), \\ y(t) &= \boldsymbol{C}h(t), \end{aligned} \tag{1}$$

where $\boldsymbol{A} \in \mathbb{R}^{N\times N}$, $\boldsymbol{B} \in \mathbb{R}^{D\times N}$, and $\boldsymbol{C} \in \mathbb{R}^{D\times N}$. Discretization aims to convert the continuous differential equations into discrete functions, aligning the model to the input signal's sampling frequency for more efficient computation [31]. Following Refs. [37, 38], the continuous parameters $(\boldsymbol{A}, \boldsymbol{B})$ can be discretized by the zero-order hold rule with a given sample timescale $\Delta \in \mathbb{R}^{D}$:

$$\begin{aligned} \bar{\boldsymbol{A}} &= \mathrm{e}^{\Delta \boldsymbol{A}}, \\ \bar{\boldsymbol{B}} &= (\mathrm{e}^{\Delta \boldsymbol{A}} - \Delta \boldsymbol{I})\boldsymbol{A}^{-1}\boldsymbol{B}, \\ \bar{\boldsymbol{C}} &= \boldsymbol{C}, \end{aligned} \tag{2}$$

$$\begin{aligned} \bar{\boldsymbol{B}} &\approx (\Delta \boldsymbol{A})(\Delta \boldsymbol{A})^{-1}\boldsymbol{A}\boldsymbol{B} = \Delta \boldsymbol{B}, \\ h(t) &= \bar{\boldsymbol{A}}h(t-1) + \bar{\boldsymbol{B}}x(t), \\ y(t) &= \bar{\boldsymbol{C}}h(t), \end{aligned}$$

where $\bar{\boldsymbol{A}} \in \mathbb{R}^{N\times N}$, $\bar{\boldsymbol{B}} \in \mathbb{R}^{D\times N}$ and $\bar{\boldsymbol{C}} \in \mathbb{R}^{D\times N}$.

To simplify calculations, the repeated application of Eq. (2) can be efficiently performed simultaneously using a global convolution approach:

$$\begin{aligned} &\boldsymbol{y} = \boldsymbol{x} * \bar{\boldsymbol{K}}, \\ \text{with} \quad &\bar{\boldsymbol{K}} = (\boldsymbol{C}\bar{\boldsymbol{B}}, \boldsymbol{C}\overline{\boldsymbol{A}\boldsymbol{B}}, \ldots, \boldsymbol{C}\bar{\boldsymbol{A}}^{L-1}\bar{\boldsymbol{B}}), \end{aligned} \tag{3}$$

where $*$ denotes the convolution operation, and $\bar{\boldsymbol{K}} \in \mathbb{R}^{L}$ is the SSM kernel.

3.1.2 Selective state space models (*S6*)

Mamba [9] improves the performance of SSM by introducing selective state space models (S6). This allows the continuous parameters to vary with the input, thereby enabling selective information processing across sequences and extending the discretization process via a selection mechanism:

$$\begin{aligned} \bar{\boldsymbol{B}} &= s_{\boldsymbol{B}}(\boldsymbol{x}), \\ \bar{\boldsymbol{C}} &= s_{\boldsymbol{C}}(\boldsymbol{x}), \\ \Delta &= \tau_{\boldsymbol{A}}(\text{Parameter} + s_{\boldsymbol{A}}(\boldsymbol{x})), \end{aligned} \tag{4}$$

where $s_{\boldsymbol{B}}(\boldsymbol{x})$ and $s_{\boldsymbol{C}}(\boldsymbol{x})$ are linear functions that project input $\boldsymbol{x}$ into an N-dimensional space, while $s_{\boldsymbol{A}}(\boldsymbol{x})$ broadens a D-dimensional linear projection to the necessary dimensions. In terms of visual tasks, VMamba introduces the 2D selective scan (SS2D) [10], which maintains the integrity of 2D image structures by scanning four directed feature sequences. Each sequence is processed independently within an S6 block and then combined to form a comprehensive 2D feature map.

3.1.3 Selective scan mechanism

In S6, matrices $\bar{\boldsymbol{B}} \in \mathbb{R}^{B\times L\times N}$, $\bar{\boldsymbol{C}} \in \mathbb{R}^{B\times L\times N}$, and $\Delta \in \mathbb{R}^{B\times L\times D}$ are derived from the input data $\boldsymbol{x} \in \mathbb{R}^{B\times L\times D}$, where B denotes the batch size. This implies that S6 is aware of the contextual information embedded in the input, ensuring the dynamism of the weights within this mechanism. VMamba introduces a cross-scan module (CSM), as depicted in Fig. 2, which maintains the integrity of the two-dimensional image structure by scanning four directed feature sequences. Each sequence is processed independently within an S6 block and then combined to form a comprehensive 2D feature map. The integration of S6 with CSM, referred to as the S6 block, serves as the core element to construct the visual state space (VSS) block.

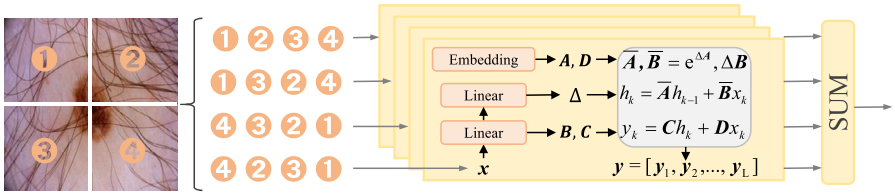


Figure 2 Visual display of the internal modeling process of selective scan mechanism [10] on an image. Here, $\boldsymbol{x}$ and $\boldsymbol{y}$ represent the input and output sequences, respectively, $\boldsymbol{y}_{\mathrm{L}}$ denotes the final element of the output sequence $\boldsymbol{y}$, and h denotes the hidden state variable. $\boldsymbol{A}, \boldsymbol{B}, \boldsymbol{C}$, and $\boldsymbol{D}$ represent the learnable parameters, while Δ is the discretization parameter used to derive $\bar{\boldsymbol{A}}$ and $\bar{\boldsymbol{B}}$. The SUM indicates the element-wise summation of the features obtained from distinct scanning directions

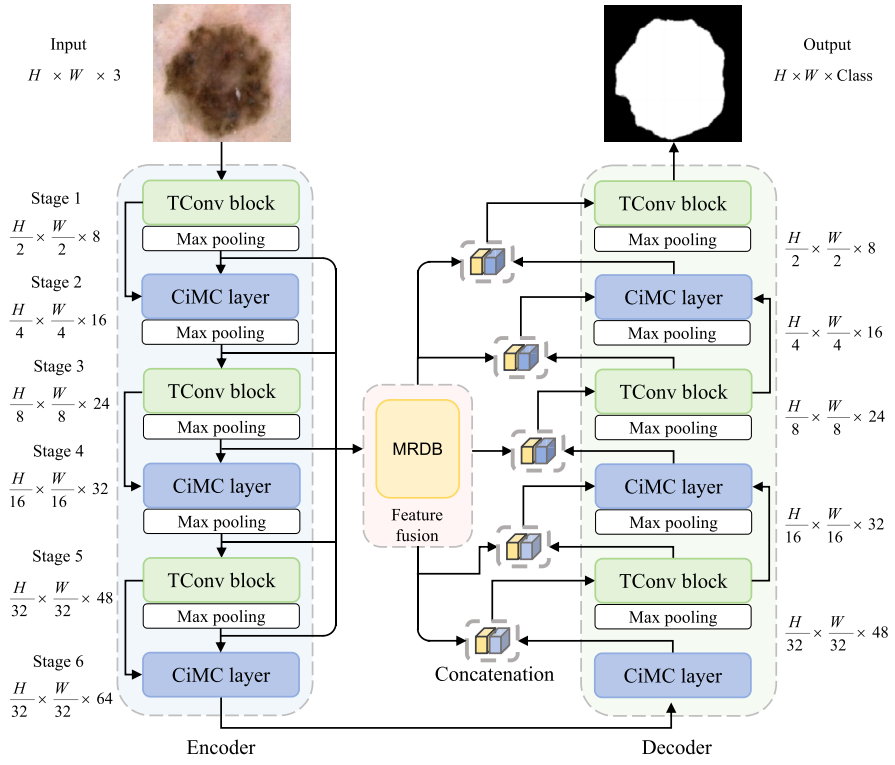


Figure 3 Framework of the proposed CIHM for medical image segmentation. The encoder primarily consists of alternately arranged T-shape convolution (TConv) blocks, context-insight Mamba-CNN (CiMC) layers, and max pooling layers. In contrast, the decoder mainly comprises CiMC layers, TConv blocks, and bilinear interpolation layers. The multiscale refining detail bridge (MRDB) module is introduced in skip connections to enable multiscale feature fusion and propagation. Here, H and W denote the height and width of the images and feature maps, respectively

3.2 Architecture overview

In this section, the proposed context-insight hybrid Mamba-CNN model (CIHM) is examined. As illustrated in Fig. 3, CIHM is structured with a symmetrical U-shaped architecture. Focusing on the encoder, it comprises six stages with channel numbers set to [8, 16, 24, 32, 48, 64]. These stages alternately employ our newly developed TConv block and CiMC layer to extract local and global features, respectively. The TConv block is formed by concatenating a partial convolution (PConv) and a point-wise convolution (PWConv) layer, which can better recover spatial textures or boundary structures. The CiMC layer consists of duplex gated fusion (DGF) and V^2SSM, enabling effective capture and interaction between local features and multi-level global features. The decoder mirrors the encoder's configuration. The skip connection pathway employs the multiscale refining detail bridge (MRDB) module for multi-level, multi-scale feature fusion [39], enhancing the decoder's capabilities. By integrating these advanced modules, CIHM significantly reduces parameters

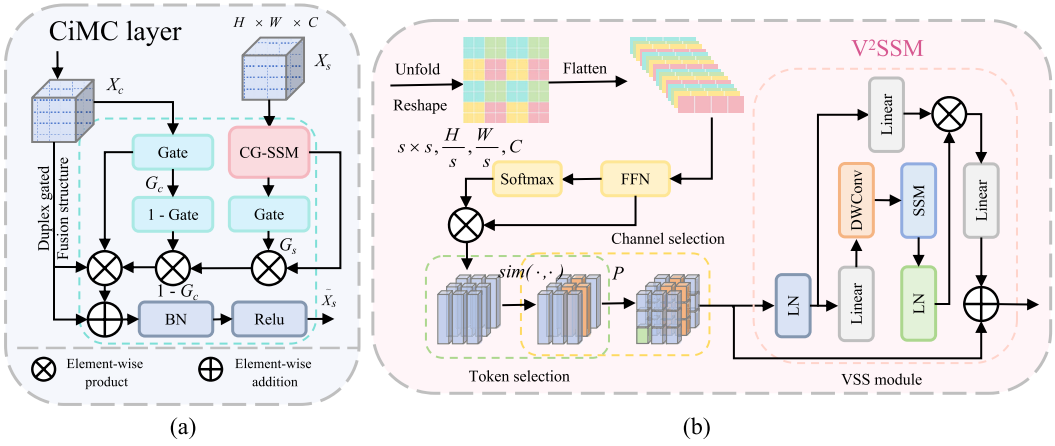


Figure 4 Detailed architectures of the core components in the proposed CIHM. (a) Detailed structure of the CiMC layer. The main component of this module is the duplex gated fusion structure, which enables the interaction and fusion of local and global features by controlling the gating logic of information flow. Here, X_c and X_s denote the input features from the CNN branch and SSM branch, respectively. G_c and G_s represent the gating factors, while BN and ReLU denote batch normalization and rectified linear unit. (b) Vision-selective selective scan mechanism (V^2SSM). Based on the visual response of the image, V^2SSM adaptively filters and weights tokens or channels, prioritizing the transmission of global information closely related to the current visual context to obtain high-quality global semantic features at low cost. FFN, LN, and DWConv represent feed-forward network, layer normalization, and depth-wise convolution, respectively. Furthermore, H, W, and C represent the height, width, and channel dimensions of the feature map, while $s \times s$ denotes the patch size used in the unfold operation, resulting in $\frac{H}{s}$ and $\frac{W}{s}$ spatial dimensions. P indicates the selected informative tokens retained for subsequent processing

and computational burden while improving segmentation performance compared to previous methods.

3.3 Context-insight Mamba-CNN layer

In medical image segmentation, capturing both local and global features is crucial for enhancing the model's ability to precisely segment lesions. Additionally, adaptive global information processing helps the model more accurately understand the relative positions and relationships between structures in images—especially when image quality is suboptimal or the contrast between the target region and background is indistinct. Moreover, this adaptive global information processing provides more targeted semantic support for local features, further facilitating accurate boundary delineation. As depicted in Fig. 4(a), we developed the CiMC layer by incorporating V^2SSM into the deep feature processing pipeline to better address these challenges.

3.3.1 Duplex gated fusion structure

The CiMC layer employs a duplex gated fusion strategy to achieve precise feature aggregation. The essence of the gating mechanism lies in measuring the usefulness of each feature vector in the feature map and dynamically adjusting the manner in which information is aggregated based on this measurement. Specifically, DGF realizes feature interaction by controlling the gating logic of information flow: Given the global and local features from previous layers, denoted as $\boldsymbol{X}_s \in \mathbb{R}^{H' \times W' \times C'}$ and $\boldsymbol{X}_c \in \mathbb{R}^{H' \times W' \times C'}$, respectively (where H', W', and C' represent the height, width, and number of channels of the intermediate feature maps), each convolutional level c is associated with a gating map $\boldsymbol{G}_c \in [0,1]^{H_c \times W_c}$. Based on these gating maps, the additive fusion form is defined as follows:

$$\tilde{\boldsymbol{X}}_s = (1 + \boldsymbol{G}_c) \cdot \boldsymbol{X}_c + (1 - \boldsymbol{G}_c) \cdot \boldsymbol{G}_s \cdot \boldsymbol{X}_s, \tag{5}$$

where the gating map $\boldsymbol{G}_c = \text{sigmoid}(w_c * \boldsymbol{X}_c)$ is estimated by a convolutional layer parameterized as $w_c \in \mathbb{R}^{1 \times 1 \times C_c}$. Here, w_c denotes the learnable weights of a 1×1 convolution kernel, and C_c indicates its corresponding channel dimension. The fusion logic for the feature vector from the convolutional level c at position (x, y) is as follows: the feature of level c is fused into level s only when $\boldsymbol{G}_c(x, y)$ is large and $\boldsymbol{G}_s(x, y)$ is small. This indicates that information is transmitted when level c contains useful local features that are absent in level s.

Through this gating mechanism, useful detailed information can be accurately adjusted to the required positions, while useless information is effectively suppressed at both the sender and receiver, thereby avoiding information redundancy (local information is received only when the current position lacks effective features). The results of these branches are then aggregated to facilitate inter-branch information exchange. The final output $\tilde{\boldsymbol{X}}_s \in \mathbb{R}^{H' \times W' \times C'}$ is obtained after applying the rectified linear unit (ReLU) and batch normalization (BN).

3.3.2 V^2SSM

Global features reveal the semantic structures of images across different scales and perspectives, which can significantly enhance the model's discriminative ability. To strengthen the modeling capability of global semantic features and achieve vision-specific adaptability, we introduce the vision-selective selective scan mechanism (V^2SSM) in the CiMC layer. This mechanism adaptively filters and weights tokens or channels based on the visual responses of images, prioritizing the transmission of global information closely related to the current visual context, thereby achieving high-quality global semantic representations at low cost.

First, we use efficient operations such as unfolding and reshaping to partition the input into spatially contiguous patches $(s \times s, H'/s, W'/s, C)$. Subsequently, channel-wise averaging is performed to compress spatial redundancy, followed by linear transformations via a feed-forward network (FFN) to model inter-channel dependencies. Next, an activation function is applied to compute spatial probability distributions, adjusting feature weights to emphasize regions rich in visual information. For the weighted outcomes, feature selection [40] is used to identify relevant features from tokens and channels. Specifically, each output token t_i from dimension $d = \frac{H' \times W'}{s \times s}$ is processed through feature selection, which weights tokens based on their cosine similarity $\mathrm{sim}(t_i, \boldsymbol{\xi})$ (range $[0, 1]$) to the context-aware embedding vector $\boldsymbol{\xi}$. The weighted token is calculated as follows:

$$\hat{t}_i = \boldsymbol{P} \cdot \mathrm{sim}(t_i, \boldsymbol{\xi}) \cdot t_i, \tag{6}$$

where $\boldsymbol{\xi} \in \mathbb{R}^{C'}$ is the context-aware embedding vector, and $\boldsymbol{P} \in \mathbb{R}^{C' \times C'}$ is the context-aware specific linear transformation matrix. Each t_i then undergoes linear transformation for channel selection, further filtering out vision-irrelevant channels. Finally, the feature map is reconstructed through reshaping and interpolation operations, yielding the final vision-optimized global features.

3.3.3 VSS module

Drawing on insights from Ref. [10], we employ the VSS module (as depicted in Fig. 4(b)) to capture spatial long-range dependencies. Initially, the module processes its input through layer normalization before diverging into two parallel branches. In the first branch, the feature channels are expanded to a size determined by λ, a predefined expansion factor, followed by activation using the SiLU function. The second branch expands the feature channels via a linear layer, applies depthwise convolution (DWConv) and SiLU activation [41], and continues with additional feature extraction through the SSM and normalization by LayerNorm. The outputs from both branches are then combined through element-wise multiplication, followed by a linear layer to mix the features, and a residual connection is established, forming the output of the VSS block. The process is described by Eq. (7):

$$\begin{aligned} \boldsymbol{X}_1 &= \mathrm{LN}(\mathrm{SSM}(\mathrm{SiLU}(\mathrm{DWConv}(\mathrm{Linear}(\boldsymbol{X}))))), \\ \boldsymbol{X}_2 &= \mathrm{SiLU}(\mathrm{Linear}(\boldsymbol{X})), \\ \boldsymbol{X}_{\mathrm{out}} &= \boldsymbol{X} + \mathrm{Linear}\left(\boldsymbol{X}_1 \odot \boldsymbol{X}_2\right), \end{aligned} \tag{7}$$

where $\boldsymbol{X}$ denotes the input feature map to this module, and $\odot$ represents the Hadamard product.

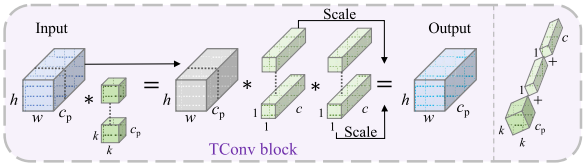


Figure 5 Detailed structure of the TConv block. It consists of partial convolution (PConv) and cascaded point-wise convolutions (PWConv). This structure forms a T-shaped receptive field that focuses on the patch center, and incorporates residual concatenation and an adjustment factor to enhance information acquisition. Here, k denotes the spatial kernel size of the convolution, while c_p and c represent the number of channels involved in the partial convolution and point-wise convolutions, respectively

3.4 TConv block

The feature maps between different channels exhibit a high degree of similarity, and this redundancy has also been reflected in several other studies [42, 43]. To effectively and efficiently exploit this redundancy, inspired by Ref. [44], we introduce the T-shaped convolution, which is a series of convolutional layers consisting of a partial convolution (PConv) followed by two pointwise convolution (PWConv) layers in tandem. As illustrated in Fig. 5, specifically, PConv is applied only to a subset of the input channels for spatial feature extraction. The remaining channels are left unchanged rather than being removed from the feature map. The FLOPs and memory access of PConv are much smaller, i.e.,

$$\begin{aligned} &h \times w \times k^2 \times c_p^2, \\ &h \times w \times 2c_p + k^2 \times c_p^2 \approx h \times w \times 2c_p. \end{aligned} \tag{8}$$

Here, h and w denote the spatial height and width of the feature maps, k represents the spatial size of the convolution kernel, and c_p indicates the number of channels involved in the partial convolution. With a typical partial ratio $r = \frac{c_p}{c} = \frac{1}{4}$ (where c is the total number of channels), the FLOPs of a PConv is only $\frac{1}{16}$ of a regular Conv, and the memory access is only $\frac{1}{4}$ of a regular Conv. The subsequent PWConv layers then optimally processes informa-

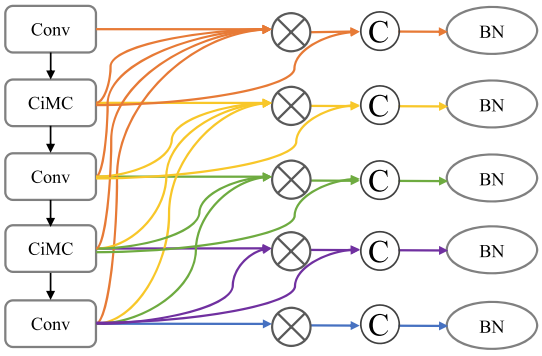


Figure 6 Detailed structure of the multiscale refining detail bridge (MRDB) module. Features are combined using multiplicative and concatenative dense connections. Composite connections represent simple operations, namely upsampling and 1 × 1 convolution

tion from all channels. Together, these layers create a T-shaped effective receptive field on the input feature map, concentrating more on the central position compared to a standard convolution, which uniformly processes a patch (see the right side of Fig. 5). T-shaped convolution not only exploits the redundancy among filters but also reduces computational and memory demands. Additionally, advanced residual concatenation and an adjustment factor are employed to enhance remote spatial information acquisition capabilities [16] and facilitate feature reuse. This configuration yields three distinct outputs, which are then combined to form the final output of the serial convolutions.

3.5 Multiscale refining detail bridge module

In medical image segmentation tasks, small lesion areas or anatomical landmarks may lose detail and boundary information in deep feature maps, making precise localization challenging. Studies [45, 46] address this by integrating shallow and deep features, but while shallow feature maps offer detailed boundary information, they also introduce noise that affects the merged image quality. Inspired by skip connection paths [39] and illustrated in Fig. 6, we introduce the MRDB, which refines image features from various scales and guides the network to efficiently combine shallow and deep features, enhancing feature perception and reducing noise.

The use of MRDB allows for fusion and interaction between multi-stage feature mappings of different scales and higher-level semantic features. Specifically, the output of the encoder is defined as $\boldsymbol{M}_t$ $(t = 1, \ldots, 5)$, and the output $\boldsymbol{N}_t$ of the feature fusion stage t is the product of the encoder's output at stage t and the encoder outputs of the remaining stages from $5 - t$, concatenated with the encoder output of the next stage. This is formalized as follows:

$$\boldsymbol{N}_t = \text{concat}\left(\prod_{i=t}^{5} f_i(\boldsymbol{M}_i), f_{i+1}(\boldsymbol{M}_{i+1})\right), \quad t \geq 1, \tag{9}$$

where $f(\cdot)$ denotes a transformation using a 1×1 convolution to reduce channels and upsampling for resolution restoration. Subsequently, we examine why CIHM, as opposed to simple additive operations without feature fusion, more effectively bridges the gap between low-level details and high-level semantics. For instance, in the third stage, the gradient computation during backpropagation is given by Eq. (10):

$$\boldsymbol{N}_3 = \text{concat}(f_3(\boldsymbol{M}_3) \cdot f_4(\boldsymbol{M}_4) \cdot f_5(\boldsymbol{M}_5), f_4(\boldsymbol{M}_4)), \tag{10}$$

$$\frac{\partial \boldsymbol{N}_3}{\partial f_3(\boldsymbol{M}_3)} = \text{concat}(f_4(\boldsymbol{M}_4) \cdot f_5(\boldsymbol{M}_5), 0). \tag{11}$$

As indicated by Eq. (9), the network's branch gradients are not fixed. If one branch fails to capture superior features, it will be influenced by the other branches, increasing the gradient magnitude. Unlike the skip connection method, which introduces noise from shallow layers, our fusion method significantly constrains the branches during training. This allows them to mutually promote and refine each other, thereby filtering out noise and achieving more accurate prediction results. Additionally, feature maps are preserved through concatenation at each stage, maintaining essential details for subsequent stages. This not only enhances the contextual relationship between different feature maps but also boosts the model's capability to represent features, thus improving overall performance.

4 Experiments

4.1 Dataset

To validate the competitive performance of the presented CIHM while maintaining low parameter and computational costs, we conducted comprehensive experiments across multiple medical image segmentation datasets. These include segmentation of skin lesions, cells, and organs. The datasets were selected to span various resolutions and imaging modalities, offering insights into the model's effectiveness and adaptability in diverse medical imaging scenarios.

ISIC2017 and ISIC2018 datasets The International Skin Imaging Collaboration's 2017 and 2018 challenge datasets (ISIC2017 and ISIC2018) [47, 48] are two large datasets for skin lesion segmentation. Specifically, the ISIC2017 dataset comprises 2000 dermatoscopic images with segmentation mask labels, of which 1250 are used for training, 150 for validation, and 600 for testing. For the ISIC2018 dataset, we obtained 2594 dermatoscopic images with segmentation masks, distributed as 1815 for training, 259 for

Table 1 Comparative experimental results on the ISIC2017 and ISIC2018 dataset. Bold indicates the best. (1) and (0) indicate that CIHM is significantly superior or not significantly different from the baseline, respectively ($p < 0.05$ via Wilcoxon signed-rank test)

Type	Method	Params (M) ↓	FLOPs (G) ↓	Time (s) ↓	ISIC2017				ISIC2018			
					DSC ↑	SE ↑	SP ↑	ACC ↑	DSC ↑	SE ↑	SP ↑	ACC ↑
CNN	U-Net	7.770	13.76	0.023	86.99±1.1$^{(1)}$	86.82±1.3$^{(1)}$	97.43±0.6$^{(1)}$	95.65±0.7$^{(1)}$	87.55±1.0$^{(1)}$	85.86±1.4$^{(1)}$	96.69±0.8$^{(1)}$	94.05±0.9$^{(1)}$
	SCR-Net	0.801	1.567	0.027	88.98±0.9$^{(1)}$	84.97±1.5$^{(1)}$	**98.53**±0.4$^{(0)}$	95.88±0.6$^{(1)}$	88.86±0.8$^{(1)}$	88.92±1.1$^{(1)}$	97.14±0.7$^{(1)}$	95.47±0.8$^{(1)}$
	C^2SDG	22.001	7.972	0.028	89.38±0.8$^{(1)}$	88.59±1.2$^{(1)}$	97.65±0.7$^{(1)}$	95.88±0.5$^{(1)}$	88.06±0.9$^{(1)}$	89.70±1.0$^{(0)}$	96.43±0.9$^{(1)}$	95.06±0.7$^{(1)}$
Hybrid Trans	TransFuse	26.160	11.50	0.084	88.40±0.7$^{(1)}$	87.14±1.1$^{(1)}$	97.98±0.5$^{(1)}$	96.17±0.4$^{(1)}$	89.27±0.6$^{(1)}$	**91.28**±0.9$^{(0)}$	95.74±0.8$^{(1)}$	94.66±0.6$^{(1)}$
	UTNetV2	12.800	15.50	0.045	87.23±0.9$^{(1)}$	84.45±1.4$^{(1)}$	98.05±0.6$^{(1)}$	95.84±0.5$^{(1)}$	88.25±0.8$^{(1)}$	87.60±1.2$^{(1)}$	96.48±0.7$^{(1)}$	94.32±0.8$^{(1)}$
Hybrid SSM	VM-UNet	27.427	4.112	0.035	90.70±0.6$^{(1)}$	88.37±0.9$^{(1)}$	98.42±0.4$^{(0)}$	96.45±0.5$^{(1)}$	88.91±0.7$^{(1)}$	88.09±1.1$^{(1)}$	97.44±0.6$^{(1)}$	95.60±0.5$^{(1)}$
	MedMamba	14.457	2.265	0.072	88.96±0.8$^{(1)}$	88.24±1.0$^{(1)}$	97.62±0.6$^{(1)}$	95.83±0.6$^{(1)}$	88.35±0.9$^{(1)}$	89.74±1.0$^{(0)}$	97.65±0.5$^{(1)}$	95.55±0.6$^{(1)}$
	LightM-UNet	0.403	0.391	0.034	90.80±0.5$^{(1)}$	88.39±0.8$^{(1)}$	98.46±0.4$^{(0)}$	96.49±0.4$^{(1)}$	88.98±0.6$^{(1)}$	88.29±1.1$^{(1)}$	97.65±0.5$^{(1)}$	95.55±0.6$^{(1)}$
	CIHM	**0.194**	**0.182**	**0.022**	**91.40**±0.38	**89.19**±0.45	98.27±0.28	**97.26**±0.32	**90.39**±0.42	89.39±0.48	**97.91**±0.30	**96.77**±0.35

validation, and 520 for testing. We performed a detailed evaluation on these two datasets, including metrics such as Dice similarity coefficient (DSC), accuracy (ACC), sensitivity (SE), and specificity (SP).

2018 Data Science Bowl (DSB18) DSB18 originates from the 2018 Kaggle competition [49]. For our experiments, we utilized the dataset's 670 annotated images. Consistent with prior research [21], the training and testing datasets were randomly divided in a 4:1 ratio. For this dataset, we employed intersection over union (IoU) and F1-score as the evaluation metrics.

Synapse multi-organ segmentation dataset Synapse [50] is an openly available dataset for multi-organ segmentation, comprising 3779 axial clinical CT images from 30 abdominal CT cases. These images include segmentation of eight abdominal organs (aorta, gallbladder, left kidney, right kidney, liver, pancreas, spleen, and stomach). Following the setup from previous work [26, 27], 18 cases were used for training and 12 for testing. For this dataset, we report the Dice similarity coefficient (DSC) and the 95% Hausdorff distance (HD95) as evaluation metrics.

4.2 Implementation details

All experiments were implemented based on Python 3.8 and PyTorch 1.13.0. We applied consistent data augmentation across experiments to fairly assess model performance, including horizontal and vertical flips, and random rotations. All methods were trained from scratch without pre-trained weights under identical experimental settings. For skin lesion and cell segmentation tasks, images were resized to 256 × 256 pixels, while for multi-organ segmentation, the size was set at 224 × 224 pixels. The Synapse dataset utilized the CeDice loss function, while other datasets employed the BceDice loss function. We used the AdamW optimizer [51], with a training epoch of 300, a batch size of 16, an initial learning rate of 0.001, and a minimum learning rate of 0.00001, managed by a cosine annealing learning rate scheduler. The experiments were all on a single NVIDIA RTX A6000 GPU.

4.3 Comparison with state-of-the-art methods

4.3.1 Comparison methods

State-of-the-art (SOTA) CNN-/Transformer-/SSM-based task-specific approaches were adopted for evaluation, including U-Net [18], SCR-Net [52], C^2SDG [53], TransFuse [54], UTNetV2 [55], VM-UNet [31], MedMamba [56], LightM-UNet [16], H-vmunet [57], R50 U-Net [27], R50 Att-UNet [27], R50 ViT [27], TransUNet [27], Att-UNet [24], TransNorm [58], SwinUNet [26], MT-UNet [59], FCN [60], ResU-Net [61], and X-Net [21].

4.3.2 Quantitative comparison

To validate the competitive performance of CIHM at low computational costs, we conducted comparative experiments with several classical medical image segmentation and lightweight models. As listed in Table 1 and Table 2, we report the mean ± standard deviation for all models to demonstrate stability. To rigorously evaluate the performance gains, we performed a Wilcoxon signed-rank test. The "1" and "0" markers in the tables denote whether the improvement in CIHM compared to the baseline is statistically significant ($p < 0.05$). For the ISIC2017 and ISIC2018 datasets, our CIHM outperforms other models across multiple metrics such as DSC and ACC. Regarding inference speed, CIHM demonstrates superior efficiency with 0.022 second per image. Compared to the well-known U-Net [18], CIHM not only demonstrated outstanding and highly competitive performance but also significantly reduced parameters and computational costs by 40 times and 76 times, respectively. Table 2 demonstrates that for the Synapse dataset, CIHM also maintained competitive performance. Notably, our model surpasses TransUNet, the pioneer in integrating Transformer with U-Net, improving DSC by 3.69% and reducing HD95 by 9.53 millimeters. As presented in Table 3,

Table 2 Comparative experimental results on the Synapse multi-organ CT dataset. DSC of every single class is also reported

Type	Method	Average		Aorta	Gall-bladder	Kidney (L)	Kidney (R)	Liver	Pancreas	Spleen	Stomach
		DSC ↑	HD ↓								
CNN	U-Net	76.85±0.9$^{(1)}$	39.70±3.2$^{(1)}$	89.07	**69.72**	77.77	68.60	93.43	53.98	86.67	75.58
	R50 U-Net	74.68±1.2$^{(1)}$	36.87±2.8$^{(1)}$	87.47	66.36	80.60	78.19	93.74	56.90	85.87	74.16
	R50 Att-UNet	75.57±1.1$^{(1)}$	36.97±2.9$^{(1)}$	55.92	63.91	79.20	72.71	93.56	49.37	87.19	74.95
	Att-UNet	77.77±0.8$^{(1)}$	36.02±2.5$^{(1)}$	**89.55**	68.88	77.98	71.11	93.57	58.04	87.30	75.75
Hybrid Trans	R50 ViT	71.29±1.5$^{(1)}$	32.87±3.5$^{(1)}$	73.73	55.13	75.80	72.20	91.51	45.99	81.99	73.95
	TransUNet	77.48±0.7$^{(1)}$	31.69±2.1$^{(1)}$	87.23	63.13	81.87	77.02	94.08	55.86	85.08	75.62
	TransNorm	78.40±0.6$^{(1)}$	30.25±2.0$^{(1)}$	86.23	65.10	82.18	78.63	94.22	55.34	89.50	76.01
	SwinUNet	79.13±0.5$^{(1)}$	21.55±1.8$^{(0)}$	85.47	66.53	83.28	79.61	94.29	56.58	**90.66**	76.60
	MT-UNet	78.59±0.6$^{(1)}$	26.59±2.3$^{(1)}$	87.92	64.99	81.47	77.29	93.06	59.46	87.55	76.81
Hybrid SSM	VM-UNet	81.08±0.4$^{(0)}$	**19.21**±1.4$^{(0)}$	86.40	69.41	86.16	**82.76**	94.17	58.80	89.51	81.40
	H-vmunet	81.13±0.4$^{(0)}$	20.56±1.6$^{(0)}$	86.85	68.12	86.55	82.10	94.25	58.95	89.20	81.35
	LightM-UNet	80.52±0.5$^{(1)}$	21.24±1.7$^{(0)}$	86.54	65.45	87.25	81.74	**94.37**	59.06	88.59	81.13
	CIHM	**81.17**±0.3	22.16±1.5	87.31	67.78	**87.71**	81.85	93.30	**60.03**	89.48	**81.91**

Table 3 Comparative experimental results on the DSB18 dataset

Type	Method	DSB18	
		IoU ↑	F1-score ↑
CNN	FCN	74.08	85.03
	U-Net	80.31	89.03
	ResU-Net	80.28	89.01
Hybrid Trans	TransUNet	81.79	89.92
	X-Net	82.50	90.36
Hybrid SSM	VM-UNet	83.92	91.15
	H-vmunet	84.41	91.58
	CIHM	**86.03**	**92.49**

for the DSB18 dataset, our approach achieved a significant lead.

4.3.3 Qualitative comparison

Figure 7 displays a visual comparison of the segmentation results across four diverse datasets. This indicates that our method provides superior qualitative performance, producing smoother and more precise object boundaries than other baseline models. To provide a more intuitive comparison, we visualized the segmentation results and analyzed the metric distributions. As depicted in Fig. 8, the visualization results clearly demonstrate that CIHM provides smoother and more precise contours and boundaries. This is attributed to our model's strong global contextual insights, which effectively address the issue of similarities between normal and lesion areas, thereby significantly enhancing segmentation precision. Furthermore, we plotted the distribution of DSC in Fig. 9, which illustrates the distribution of prediction counts across different DSC scores. The purple histogram of CIHM has a larger proportion of samples concentrated in the high DSC interval, reflecting the superior consistency of CIHM across individual test samples and indicating fewer failure cases.

4.3.4 Robustness analysis

Given that noise contamination is a prevalent issue in medical imaging, we employed the widely used additive Gaussian noise model. We assessed the performance of the proposed method when the source images were contaminated with Gaussian noise at different intensities: $\sigma = 0.1$, $\sigma = 0.15$ and $\sigma = 0.2$. Detailed results are presented in Table 4. A general performance degradation is observed with increasing levels of noise. Notably, our method demonstrates consistently superior performance, demonstrating its superior noise resilience. Additionally, we evaluated robustness to small lesions by testing on the top 5% smallest samples. CIHM achieved a DSC of 85.64% on this challenging subset, verifying its capability to capture fine-grained details.

4.4 Ablation study

We conducted a series of ablation experiments on the ISIC2017 dataset to verify the effectiveness of our presented modules. Initially, we performed a macro ablation analysis to evaluate the performance contributions of the CiMC layer, TConv block, and MRDB module in our model. We replaced the CiMC layer and TConv block with standard convolutional blocks from other stages in sequence. For the skip connections, we substituted the MRDB module with standard U-Net concatenative fusion and a simpler additive fusion strategy, respectively. As summarized in Table 5, the decline in all performance metrics strongly confirms the crucial role of our proposed

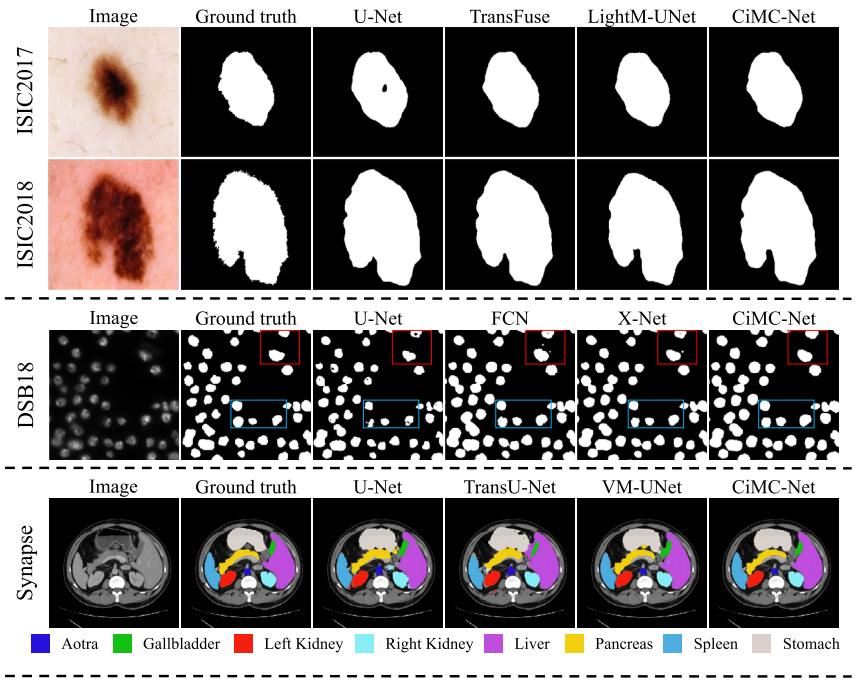


Figure 7 Visual comparison of segmentation results produced by different methods across multiple datasets, with legends for multiple organs displayed at the bottom. Our method (the last column) exhibits the smoothest boundaries and the most accurate segmentation outcomes

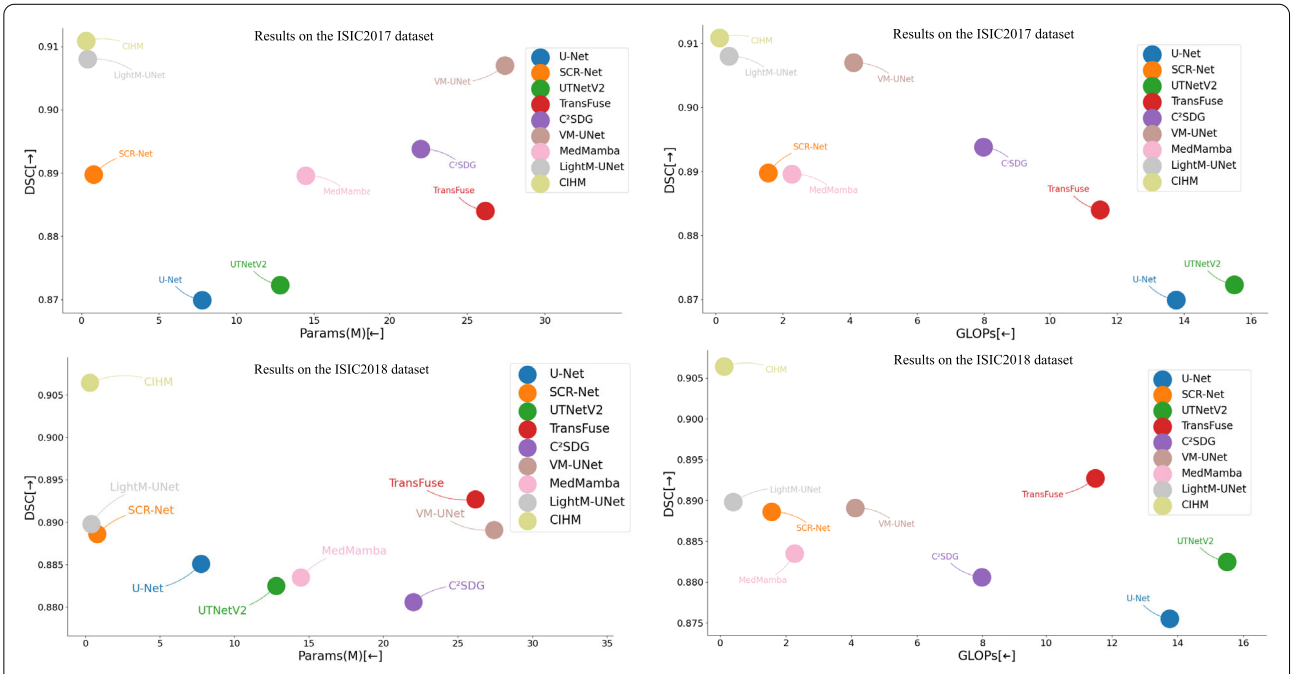


Figure 8 Visualization of comparative results on the ISIC2017 and ISIC2018 datasets. The *X*-axis corresponds to the number of parameters and GFLOPs (lower is better), while the *Y*-axis corresponds to the segmentation performance (DSC, higher is better)

modules in refining image features and facilitating the fusion and interaction of multi-scale features.

Subsequently, we carried out a detailed micro ablation study on the CiMC layer. Specifically, we sequentially re-

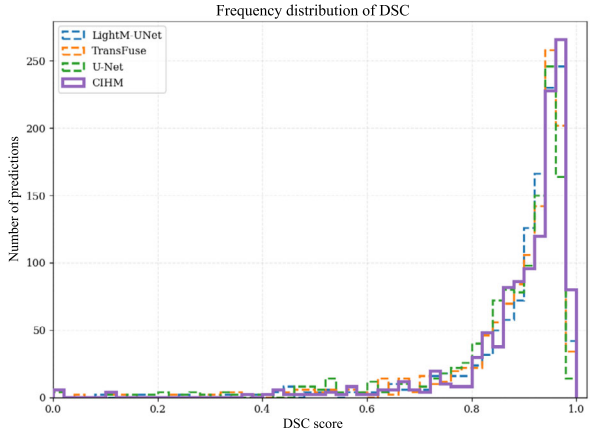


Figure 9 Frequency distribution of DSC on the ISIC2017 dataset. The horizontal axis represents the DSC, and the vertical axis indicates the number of test samples

Table 4 Comprehensive robustness evaluation on the ISIC2017 dataset. We assess the model under image degradation (Gaussian noise) and varying lesion sizes

Test scenario	DSC	Drop (Δ)	Retain (%)
Standard setting			
Clean (Overall)	**0.8729**	–	100.0
Noise robustness			
$\sigma = 0.10$	0.8705	0.0024	99.7
$\sigma = 0.15$	0.8655	0.0074	99.1
$\sigma = 0.20$	0.8580	0.0149	98.3
Lesion size sensitivity			
Large	0.8780	+0.0051	100.6
Small (Top 5%)	0.8564	0.0165	98.1

moved the input of local information and global information, and replaced the DGF with simple concatenation. To further validate the superiority of the DGF structure, we compared the CiMC layer against a standard VSS block. Our proposed approach, which integrates the advantages of CNN and Mamba, achieved excellent performance with DSC and ACC scores of 91.40% and 97.26%, respectively, significantly outperforming other methods. Furthermore, we removed token selection and channel selection from the V^2SSM. The results demonstrated that these components effectively enhanced model performance while maintaining the existing number of parameters and computational overhead, as illustrated in the visualization results of Fig. 10 and Fig. 11.

4.5 Further discussion

Although CIHM demonstrates substantial improvements, certain areas still require further refinement. First, the current 2D patch scanning paradigm in SSM branches restricts spatial continuity modeling for volumetric data, necessitating an extension to 3D selective scanning. Second, while the MRDB module significantly enhances boundary details, precision metrics such as HD95 indicate potential for further refinement in complex anatomical regions, possibly through the integration of active contour mechanisms or learnable scanning trajectories. Future work will focus on extending this framework to 3D segmentation and adaptive scanning protocols to further enhance its applicability in diverse medical scenarios.

Table 5 Ablation studies on the ISIC2017 dataset. (a) Macro ablation studies focusing on key components. (b) Micro ablation studies conducted on the CiMC layer. (c) Presentation of our method. Abbreviations: Loc. Feat. = local feature, Glo. Feat. = global feature, Tok. Sel. = token selection, Ch. Sel. = channel selection

Type	Method	DSC (%) ↑	ACC (%) ↑
(a)	w/o CiMC layer	88.76 ± 0.15	96.22 ± 0.11
	w/o TConv block	90.85 ± 0.12	96.95 ± 0.09
	Skip (Concat)	90.45 ± 0.14	96.96 ± 0.10
	Skip (Add)	90.25 ± 0.13	96.90 ± 0.12
(b)	VSS Block	90.72 ± 0.11	97.05 ± 0.09
	w/o DGF	90.18 ± 0.13	96.83 ± 0.12
	w/o Loc. Feat.	90.70 ± 0.11	97.01 ± 0.08
	w/o Glo. Feat.	91.17 ± 0.10	97.18 ± 0.07
	w/o Tok. Sel.	90.95 ± 0.12	97.07 ± 0.09
	w/o Ch. Sel.	91.01 ± 0.11	97.12 ± 0.08
(c)	CIHM (Ours)	**91.40 ± 0.09**	**97.26 ± 0.06**

5 Conclusion

This study thoroughly explores the effective integration of the strengths of the Mamba and CNN architectures, presenting CIHM: a medical image segmentation method that combines the advantages of both. We introduce the CiMC layer and the TConv block, which extract local and global information alternately. The CiMC layer efficiently captures long-range global dependencies and enables deep interaction with local features via the duplex gated fusion structure, achieving vision-adaptive global semantic understanding. In contrast, the TConv block focuses on patch centers to accurately extract local detailed features. To fully leverage multi-scale and refined image features, we develop the MRDB module. Compared with the first hybrid CNN-Mamba model U-Mamba, CIHM reduces computational costs by over 99.9%. Experimental results demonstrate that CIHM maintains excellent segmentation performance and robustness while significantly reducing resource consumption.

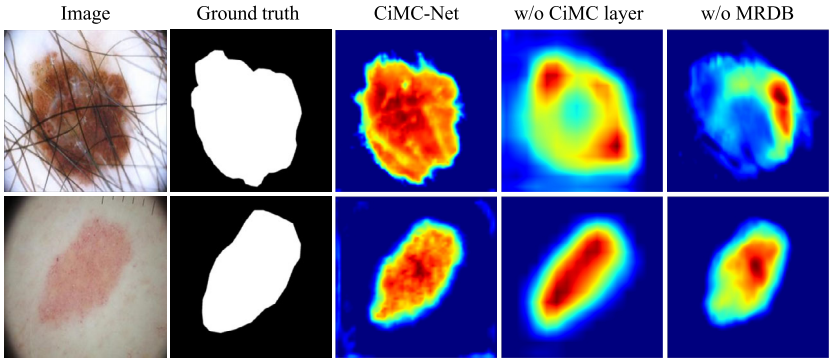


Figure 10 Comparison of feature heat maps with ablation of key model components

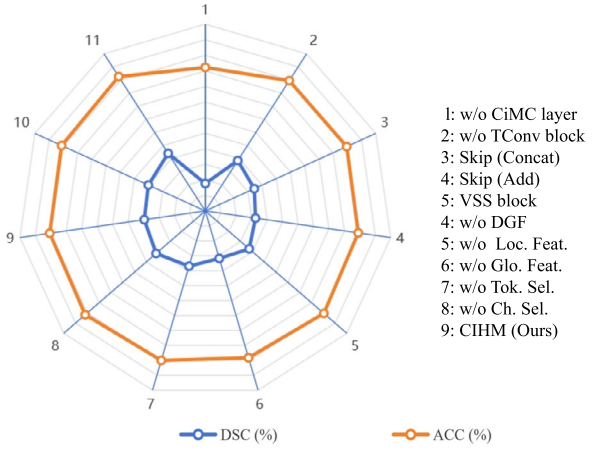


Figure 11 Visualization of detailed ablation study results. We compare segmentation accuracy between CIHM and its ablated versions

Abbreviations

CIHM, context-insight hybrid Mamba; CiMC layer, context-insight Mamba-CNN layer; CNN, convolutional neural network; DSC, dice similarity coefficient; DWConv, depthwise convolution; FLOPs, floating-point operations; HD95, 95% Hausdorff distance; IoU, intersection over union; MRDB, multiscale refining detail bridge; SSM, state space model; TConv block, T-shaped convolution block; V^2SSM, vision-selective selective scan mechanism; VSS module, visual state space module.

Author contributions

All authors contributed to the study conception and design. YM was responsible for data curation, formal analysis, investigation, methodology development, software implementation, visualization, writing the original draft, and reviewing and editing the manuscript. YL contributed to conceptualization, funding acquisition, resource provision, supervision, and review and editing of the manuscript. JC participated in methodology development, visualization, and review and editing of the manuscript. HZ was involved in investigation, validation, and review and editing of the manuscript. GQ contributed to investigation, visualization, and review and editing of the manuscript. XZ was involved in experimental implementation, validation of experimental data, and result verification. ZZ participated in conceptualization and review and editing of the manuscript. All authors read and approved the final manuscript.

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Data availability

The ISIC2017 and ISIC2018 datasets used in this paper are available at https://challenge.isic-archive.com/data/. The Data Science Bowl 2018 (DSB18) dataset is accessible at https://www.kaggle.com/c. The Synapse dataset used in this study is available at https://www.synapse.org/#!Synapse:syn3193805/wiki/217789. The source code of the proposed CIHM method will be publicly available at https://github.com/yzma/CIHM.

Declarations

Competing interests

The authors have no relevant financial or non-financial interests to disclose.

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